

TECHNICAL DOCUMENTATION — Book Framework Analyzer

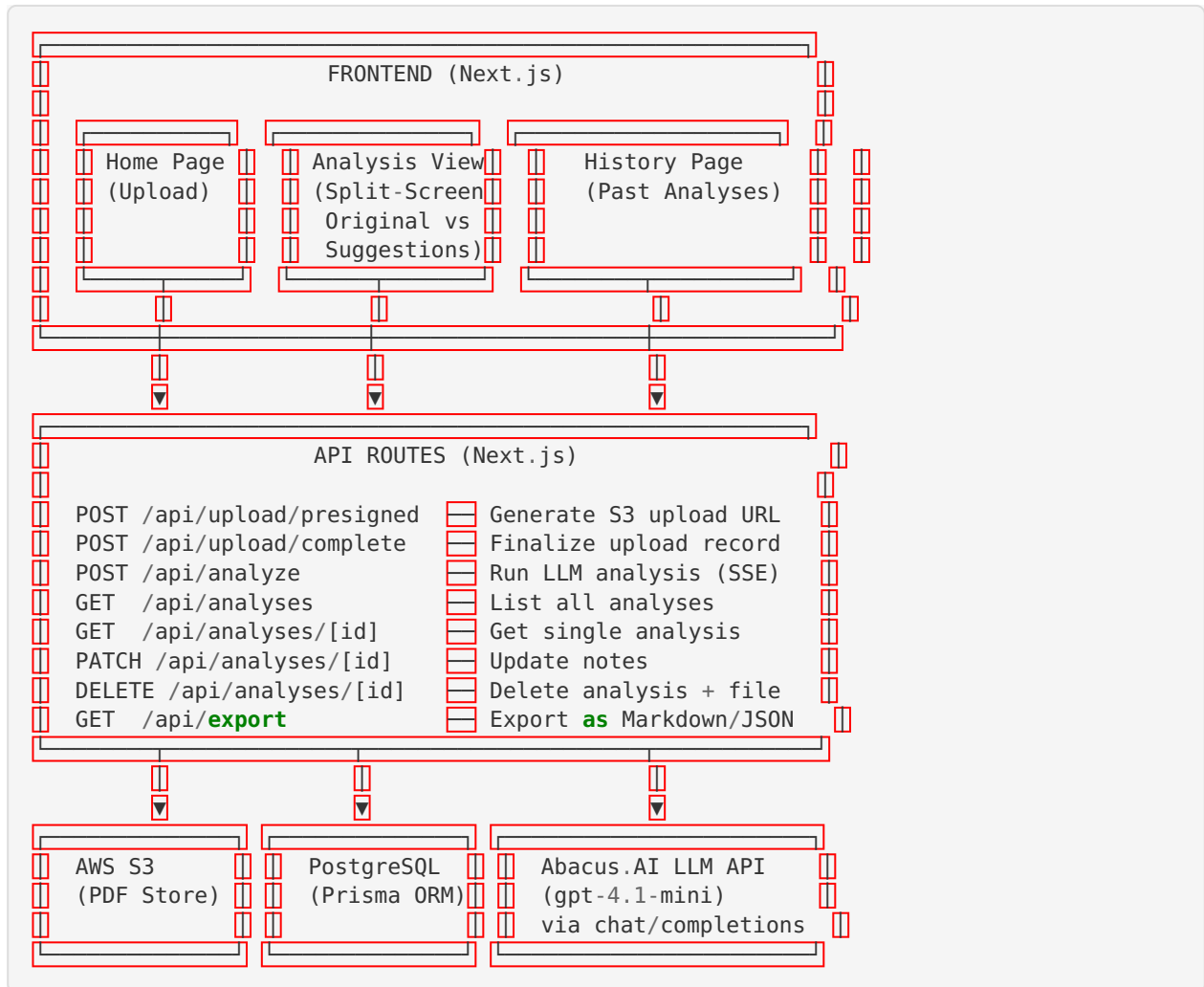
Date: April 13, 2026

Project Path: /home/ubuntu/book_framework_analyzer

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1. Architecture Overview



2. Technology Stack

Layer	Technology	Details
Frontend	Next.js 14.2.28	App Router, React Server Components
Styling	Tailwind CSS	Custom theme with dark/light mode
Animations	Framer Motion	Page transitions and UI animations
Icons	Lucide React	Consistent icon library
Toasts	React Hot Toast	User notifications
Database	PostgreSQL	Via Prisma ORM
ORM	Prisma Client	Type-safe database access
File Storage	AWS S3	Presigned URL uploads
LLM	Abacus.AI API	gpt-4.1-mini model via OpenAI-compatible endpoint
Language	TypeScript	Full type safety

3. File Structure

```

book_framework_analyzer/
├── nextjs_space/
│   ├── prisma/
│   │   ├── schema.prisma          # Database schema
│   │   └── lib/
│   │       ├── db.ts              # Prisma client singleton
│   │       ├── s3.ts              # S3 operations (presigned URLs, delete)
│   │       ├── aws-config.ts      # AWS S3 configuration
│   │       ├── framework-knowledge.ts # Bestseller framework knowledge base
│   │       ├── types.ts           # TypeScript type definitions
│   │       └── utils.ts           # Utility functions
│   └── app/
│       ├── layout.tsx             # Root layout with theme provider
│       ├── page.tsx               # Home page (upload + info)
│       ├── globals.css            # Global styles
│       ├── history/
│       │   └── page.tsx           # Analysis history page
│       ├── analysis/
│       │   ├── [id]/
│       │   │   ├── page.tsx      # Analysis detail server page
│       │   │   └── components/
│       │   │       └── analysis-detail-client.tsx # Split-screen view
│       └── api/
│           ├── upload/
│           │   ├── presigned/route.ts # Generate presigned upload URL
│           │   ├── complete/route.ts  # Complete upload + create record
│           │   ├── analyze/route.ts    # LLM analysis endpoint (SSE)
│           │   └── analyses/
│           │       ├── route.ts        # List all analyses
│           │       ├── [id]/route.ts   # CRUD single analysis
│           │       └── export/route.ts  # Export as Markdown
│           └── components/
│               ├── header.tsx         # App header with nav
│               ├── upload-zone.tsx    # Drag-and-drop PDF upload
│               ├── analysis-view.tsx  # Split-screen analysis component
│               ├── theme-provider.tsx # Dark/light mode provider
│               └── ui/                # Shadcn UI components
│               # Framework research files
│   ├── data/
│   │   ├── alltime_best sellers.md
│   │   ├── recent_best sellers.md
│   │   ├── framework_analysis.md
│   │   └── final_report.md
│   ├── .env                         # Environment variables
│   ├── next.config.js
│   ├── tailwind.config.ts
│   └── tsconfig.json
├── business_books_research/        # Original research output
│   ├── alltime_best sellers.md (+ PDF)
│   ├── recent_best sellers.md (+ PDF)
│   ├── framework_analysis.md (+ PDF)
│   └── final_report.md (+ PDF)

```

4. Database Schema

```

model Analysis {
  id          String    @id @default(cuid())
  filename    String
  cloudStoragePath String
  originalStructure Json? // Parsed chapter structure from manuscript
  suggestions  Json?     // Full LLM analysis results
  frameworkMatch String? // Primary framework match name
  notes        String?   @db.Text // User annotations
  status        String    @default("pending") // pending|analyzing|completed|error
  createdAt     DateTime @default(now())
  updatedAt     DateTime @updatedAt
}

```

Status Flow:

```

pending → analyzing → completed
                → error

```

5. API Endpoints

POST /api/upload/presigned

- **Purpose:** Generate a presigned S3 URL for PDF upload
- **Request:** { filename, contentType }
- **Response:** { presignedUrl, cloudStoragePath, headers }
- **Auth:** None (personal use)

POST /api/upload/complete

- **Purpose:** Create Analysis record after upload completes
- **Request:** { filename, cloudStoragePath }
- **Response:** { id, filename, status } (Analysis record)

POST /api/analyze

- **Purpose:** Run LLM-powered framework analysis on uploaded manuscript
- **Request:** { analysisId, chapterByChapter? }
- **Response:** Server-Sent Events (SSE) stream
- **LLM Model:** gpt-4.1-mini via Abacus.AI API
- **Process:**
 1. Retrieve PDF from S3
 2. Convert to base64
 3. Send to LLM with framework knowledge prompt
 4. Stream response back to client
 5. Save results to database on completion

Analysis JSON Structure:

```

{
  "currentStructure": {
    "chapters": [
      {
        "title": "Chapter title",
        "chapterNumber": 1,
        "content": "Extracted text (500+ words)",
        "keyPoints": ["Point 1", "Point 2"]
      }
    ],
    "totalChapters": 10,
    "estimatedWordCount": 50000
  },
  "componentScores": {
    "story": { "score": 0-100, "present": [...], "missing": [...], "fixes": [...] },
    "principles": { "score": 0-100, "present": [...], "missing": [...], "fixes": [...] },
    "system": { "score": 0-100, "present": [...], "missing": [...], "fixes": [...] },
    "research": { "score": 0-100, "present": [...], "missing": [...], "fixes": [...] }
  },
  "overallScore": 0-100,
  "prioritizedFixes": [
    {
      "priority": 1,
      "component": "story|principles|system|research",
      "issue": "Description",
      "fix": "Action to take",
      "impact": "Why it matters",
      "bestseller_example": "How X book does this"
    }
  ],
  "transformationSummary": "Overall path to bestseller-ready"
}

```

GET /api/analyses

- **Purpose:** List all saved analyses
- **Response:** Array of Analysis records

GET /api/analyses/[id]

- **Purpose:** Get single analysis with full results
- **Response:** Full Analysis record with suggestions JSON

PATCH /api/analyses/[id]

- **Purpose:** Update user notes/annotations
- **Request:** { notes }
- **Response:** Updated Analysis record

DELETE /api/analyses/[id]

- **Purpose:** Delete analysis and associated S3 file
- **Response:** { success: true }

GET /api/export

- **Purpose:** Export analysis as Markdown
- **Query Params:** id, format ("markdown"), download ("true" for attachment)

- **Response:** JSON { content, filename }

6. Frontend Pages & Components

Pages

Route	Component	Description
/	app/page.tsx	Home page with upload zone, framework cards, how-it-works
/history	app/history/page.tsx	List of all past analyses with status, scores, actions
/analysis/[id]	app/analysis/[id]/page.tsx	Individual analysis detail with split-screen view

Key Components

Component	File	Description
Header	components/header.tsx	Nav bar with branding, history link, theme toggle
UploadZone	components/upload-zone.tsx	Drag-and-drop PDF upload with progress
AnalysisView	components/analysis-view.tsx	Split-screen: original structure vs suggestions
AnalysisDetailClient	app/analysis/[id]/_components/analysis-detail-client.tsx	Full detail view with chapter navigation

Split-Screen UI Layout

LEFT PANEL	RIGHT PANEL
Original Structure	Component Scores (Story, Principles System, Research)
- Chapter titles - Key points - Content preview	Fixes & Suggestions
	- What's present - What's missing - How to fix - Bestseller examples
	Prioritized Fixes
	#1 highest impact #2 next
	Transformation Summary

7. Framework Knowledge Base

The app evaluates manuscripts against **The Principle-Based Hybrid Framework** — the only framework that has produced 100M+ combined book sales.

The 4 Essential Components

Component	What It Evaluates	Bestseller Examples
Story Elements	Personal anecdotes, case studies, emotional narratives, character-driven examples	Rich Dad Poor Dad (two fathers parable), Atomic Habits (personal injury story)
Principle Elements	Named laws/rules/habits, proprietary vocabulary, core philosophy	7 Habits (7 numbered habits), Atomic Habits (4 Laws), Think and Grow Rich (13 principles)
System Elements	Step-by-step tactics, checklists, templates, action items, before/after path	Atomic Habits (specific tactics), Deep Work (4 rules with sub-strategies)
Research Elements	Data, statistics, studies, expert citations, scientific backing	Atomic Habits (behavioral psychology), Good to Great (5 years research)

Scoring

- **Strong:** 80-100% — Well-developed throughout
- **Moderate:** 40-79% — Present but inconsistent
- **Weak:** 1-39% — Barely present
- **Missing:** 0% — Absent

Bestseller-ready = 70%+ on ALL FOUR components

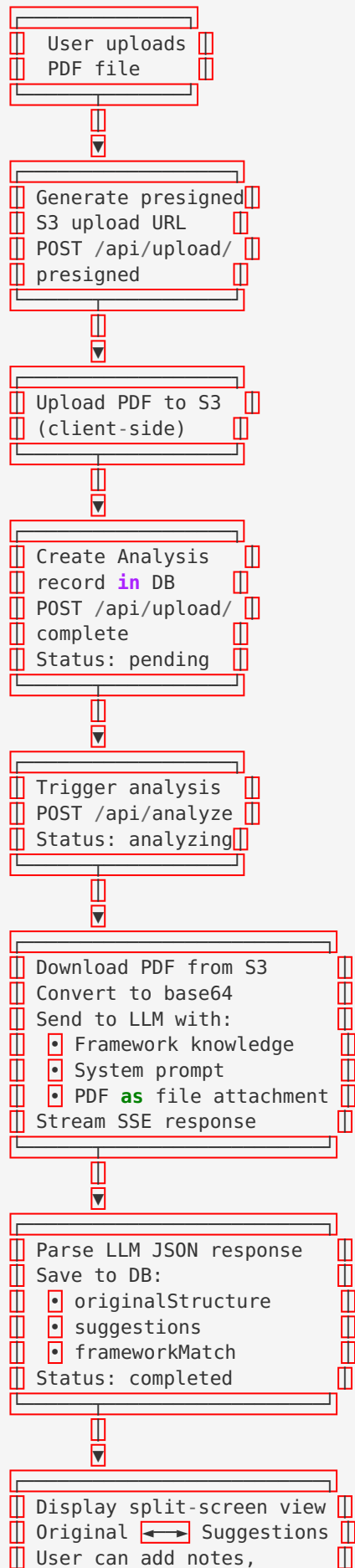
Optimal Structure Blueprint

- **Part 1: The Idea (30-40%)** — Opening hook, big idea, evidence, vision
- **Part 2: The System (60-70%)** — Framework overview + chapters per principle (story + research + tactics + action)

8. Environment Variables

Variable	Purpose
<code>DATABASE_URL</code>	PostgreSQL connection string (Prisma)
<code>AWS_PROFILE</code>	AWS credentials profile
<code>AWS_REGION</code>	S3 bucket region
<code>AWS_BUCKET_NAME</code>	S3 bucket for PDF storage
<code>AWS_FOLDER_PREFIX</code>	S3 key prefix for uploads
<code>ABACUSAI_API_KEY</code>	API key for LLM analysis

9. Analysis Workflow



```
❏ export, or re-analyze ❏
```

10. Deployment

Current Status: Built and checkpointed. NOT YET deployed to a public URL.

To Deploy: Click the Deploy button in the Abacus.AI UI, or use the `deploy_nextjs_project` tool.

Production Build:

```
cd /home/ubuntu/book_framework_analyzer/nextjs_space  
yarn run build
```

Build Output: Standalone Next.js build in `.build/standalone/`